Guilherme Trindade

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gui-portfolio-tau.vercel.app

EDUCATION

George Mason University — *January 2026 – Present*
PhD Candidate in Bioinformatics and Computational Biology, School of Systems Biology — Fairfax, VA/Hybrid
GPA: 4.0 | Relevant Coursework: AI and Deep Learning in Bioinformatics, Biological Data Analysis, Introduction to Biophysics

New York University — *September 2023 – May 2025*
M.S. in Quantitative and Computational Biology, Graduate School of Arts and Sciences — New York City, NY
Relevant Coursework: Programming for Biologists, Biological Databases & Data Mining, Applied Genomics, Immunobiology, Statistics

University of Delaware — *September 2019 – December 2022*
B.S. in Biological Sciences, College of Arts and Sciences — Newark, DE
Relevant Coursework: General Physiology, Genetics, Physics, Organic Chemistry, Microbiology, Calculus, Biological Data Analysis, Ecology

WORK EXPERIENCE

NYU Langone School of Medicine – Laboratory of Translational Obesity Research (LTOR) — *March 2024 – Sept 2025*
Graduate Research, Aleman Lab — New York City, NY

- Developed and maintained a reproducible R pipeline (FastQC → fastp/Trimmomatic → STAR → featureCounts → DESeq2 → mixOmics/LipidSearch/MetaboAnalyst) including automated QC, batch correction, normalization, and HTML reporting on NYU HPC.
- Processed human and mouse bulk and single-cell RNA-seq data in Python and R, profiling macrophage transcriptional responses to monoglyceride and triglyceride treatments, coordinated bench experiments and computational workflows.
- Integrated Claude AI into computational workflows to prototype bioinformatics pipelines, debug R/Python scripts, and develop interactive web-based data visualization tools; actively exploring agentic AI applications for multi-omics analysis.
- Sourced obesity-research papers from PubMed/bioRxiv for biweekly lab meetings; led discussion segments and shared slide decks distilling key translational and clinical insights.

Additional Prior Experience:

- NYU Langone – Dept. of Neurology (Jan 2024): memory/Alzheimer's research, Python clustering/ML, brain slicing & IHC. Univ. of Delaware – Plant & Soil Sciences (Fall 2022): Cd/Pb transport, synchrotron X-ray sample prep. Tendon Lab, STAR Campus (Summer/Fall 2022): Achilles injury ultrasound/elastography. Dialysis clinic shadowing (Neckarsulm, Germany). Lead referee, UD intramurals (100+ games).

PROJECTS

Multi-Omic Integration of Monoglyceride Response in Model Adipose-Tissue Macrophages — *March 2024 – May 2025*
Graduate Research Project – Alemán Lab, NYU Langone

- Led multi-omics investigation of macrophage inflammation in obese adipose tissue; integrated bulk/scRNA-seq and spatial/bulk LC-MS lipidomics to uncover monoglyceride interactions relevant to metabolic disease.
- Built end-to-end HPC workflow with automated QC, batch correction, RMarkdown reporting (available on GitHub); identified five MG hotspots and TG (46:9) cluster; linked PPAR signaling via GSEA/KEGG; built gene-lipid networks in Cytoscape.
- Coordinated cross-functional team (postdocs, statisticians, LC-MS engineers); generated ggplots/dashboards; mentored juniors. Poster presented at Obesity Week San Antonio (Nov 18, 2024).

Seatd — AI Front Desk SaaS for Salons, Barbershops & Nail Studios
Founder & Solo Builder

- Founded and built a full-stack SaaS automating the front desk for beauty & grooming businesses via AI-powered SMS. Handling bookings, appointment reminders, missed call recovery, Google review requests, and SMS marketing campaigns.
- Engineered 8 production n8n workflows integrating Claude Sonnet, Twilio SMS, and Google Sheets; deployed complete go-to-market stack including a landing page on Vercel, 4-step client onboarding flow, Stripe subscription billing, and automated phone number provisioning per client.

Full-Stack Sports Web Applications — *Ongoing Personal Project*

- Built two full-stack sports web apps in React 19, TypeScript, and Vite: NBA Analytics Dashboard - Express.js/Node.js backend proxying the NBA Stats API with in-memory TTL caching, Three.js/React Three Fiber 3D visuals, and Plotly.js stat charts, deployed on Railway; FIFA World Cup 2026 Fan Site – deployed on Vercel. Both open source on GitHub (github.com/noorgg22).

CERTIFICATES

New York University & HarvardX
Certificate in Software Development with Python (NYU SPS) | Data Science Professional Certificate (HarvardX)

- Mastered Python, OOP, Git & Agile; built/deployed full-stack web apps (Flask/Django); tuned ML pipelines in R/caret (k-NN, regression, ensembles); PCA, cross-validation & overfitting diagnostics for a movie recommendation system.

TECHNICAL SKILLS

AI & Machine Learning: Model building/tuning (k-NN, regularized regression/logistic, ensembles), PCA, cross-validation, recommender systems; Agentic AI & LLM tooling: Claude (Anthropic) for pipeline dev, code generation & interactive data viz (ongoing)

Programming: Python (NumPy, pandas, scikit-learn, Flask/Django), R (caret, tidyverse, Bioconductor), Bash/Linux CLI, SQL, HTML/CSS/JavaScript, Git/GitHub

Bioinformatics & Multi-omics: Bulk/scRNA-seq (DESeq2, STAR, featureCounts), lipidomics/metabolomics (LipidSearch, MetaboAnalyst, mixOmics), GSEA/KEGG, Cytoscape, NCBI/Ensembl APIs, HPC/Slurm

Tools: Jupyter, Excel (advanced), PowerPoint, JMP, BioRender, Microsoft 365, Linux administration

Languages & Interests: English (native) | Portuguese (fluent) | Spanish (conversational) | Health & Fitness, Sports Analytics, Astronomy, Gaming, Travel, Movies, Chess, Basketball